Holistic Accuracy and the Limits of Lockean Derivation

Abstract. Dorst argues that maximizing expected accuracy yields Lockean thresholds for belief even without separability. I distinguish two readings of his holistic extension. Mere truth-directedness and order independence do not suffice: a score rewarding an error-free body of belief makes a proposition believed at one credence and withheld at a higher one, with values and context fixed. The stronger reading requires restrictions on marginal values that the stated argument does not establish. For a single score, marginal extensionality and background-independent thresholds recover separability on a specified domain. A separate theorem gives cross-space refinement invariance as one sufficient route to thresholds, not a necessary condition for Lockeanism on a fixed space. The counterexample's optima select a core and its above-threshold consequences. An earlier utility theory of guessing already generates similar behavior; the contribution is its realization under the strict marginal truth-directedness at issue in Dorst's argument. Accuracy maximization alone therefore leaves the bridge from credence to belief unsettled.

1. Introduction

Accuracy-first epistemology holds that rational doxastic states are those that best promote accuracy by the agent's own lights. With a separable score for outright belief, positive for true beliefs and negative for false ones, it yields a Lockean thesis: believe a proposition just in case your credence in it exceeds a threshold. Easwaran (2016) proved this, using expected score as a tool for showing such belief states non-dominated, and stated the analogue for values varying with the proposition, which yields proposition-specific thresholds; Dorst (2019) presents these results as his Theorem 1 and Corollary 2 and credits both to Easwaran (p. 183). Dorst's own further claim is about holism. Separability, the assumption that a state's accuracy is the sum of its members' accuracies, "seems to contradict epistemic holism" (p. 192, citing Joyce 2009, §5). Dorst argues that dropping it changes nothing: given a premise he calls Sequence-Irrelevance, fixed proposition-specific thresholds follow anyway (his Proposition 5).[1]

My thesis is that Proposition 5 does not follow on the unrestricted reading of holism suggested by its motivation. On a stronger reading, the values introduced in the argument already restrict holism, and those restrictions need defending. I give a counterexample to the unrestricted claim, identify the restrictions that repair the derivation, and examine refinement invariance as a separate sufficient route to thresholds. I do not claim that Lockeanism or Separability is false, or that my counterexample is the correct measure of epistemic value.

The argument has three parts. First, a truth-directed score with a premium for an error-free belief state violates fixed thresholds for every positive premium (Section 3). The bridge from optimization to rationality requires only that some state be permitted at each credence. Second, order independence does not justify the restrictions on marginal values used in Dorst's proof (Section 4). For a single measure satisfying marginal extensionality, background-independent marginal thresholds require separability on the stated domain. Third, the score's optima have an exact core-based characterization (Section 5). Refinement invariance can recover Lockean thresholds, but its unrestricted cross-space form is stronger than the fixed-space conclusion originally sought (Section 6).

The closest prior work includes Rothschild's (2023) additive scoring representations of Lockean belief and Wang's (2025) expected-utility rules for deductively cogent belief states. Levi (1967) accepts the consequences of an answer chosen by expected epistemic utility; Leitgeb

[1] Page references to Dorst are to the published article.

(2014, 2017) connects rational belief to stably high credence and a strongest believed proposition. Holguín (2022) defends non-monotone, question-sensitive rational thinking. Dorst and Mandelkern (2022) derive filtered guesses by maximizing expected answer-value and connect belief to their consequences. Their model already reproduces the two belief sets in my main example. The claimed increment is consequently narrower: a counterexample under strict marginal truth-directedness on unrestricted belief states, a diagnosis of Dorst's derivation, and an exact characterization of this particular score's optima. Section 5 makes the overlap explicit.

The philosophical issue is the division of work between accuracy and a theory of belief's role. If a question licenses selecting a working hypothesis, an accuracy-based account can permit dependence on competing answers. If each proposition must instead be evaluated independently of those competitors, additional constraints are needed. Neither the counterexample nor its repairs decide which conception of belief is right. They show where that substantive choice enters, rather than letting an optimization theorem settle it silently.

2. Dorst's derivation

Let W be a finite set of worlds, propositions subsets of W, and a belief state B any set of propositions. Accuracy First leaves inconsistent, nonclosed and empty states among the options (Dorst 2019, p. 183). An accuracy measure assigns a real number $A(B, w)$ to each state at each world. Given a probability function C on W, expected accuracy is $EA_C(B) = \sum_w C(w)A(B, w)$, and Maximize Expected Accuracy says that B is rational only if no state has higher expected accuracy. To connect optimization to rational permission, I require only:

(Non-vacuity) At every credence under consideration, at least one belief state is rationally permitted.

Maximize Expected Accuracy alone gives only a necessary condition. Together with Non-vacuity it makes a unique optimum rationally permitted; it need not permit every tied optimum. Section 7 returns to this distinction.

Under Separability, $A(B, w) = \sum_{p \in B} v_p(w)$, where $v_p(w) = T_p > 0$ if $w \in p$ and $v_p(w) = F_p < 0$ otherwise. Then $EA_C(B) = \sum_{p \in B} e_p$ with

$$e_p = C(p)T_p + (1 - C(p))F_p,$$

so every optimum includes p when $e_p > 0$ and excludes it when $e_p < 0$. The threshold is $\tau_p = -F_p/(T_p - F_p)$. Call the values conservative when $T_p < -F_p$; then $\tau_p > 1/2$. This extends Dorst's Conservativeness, $|T| < |F|$ for uniform values (p. 185); his Variable Conservativeness for proposition-dependent values instead requires $|T_p| < |F_{\neg p}|$ and $|T_{\neg p}| < |F_p|$ (p. 191), and lets thresholds fall below $1/2$.

In §4.2 Dorst lets the measure depend on the agent's current state B_0, "to capture the sense in which it matters what other beliefs she holds" (p. 193). Adding p to B_0 raises the score where p is true and lowers it where p is false; Dorst writes the increments as $T_p^{B_0}$ and $F_p^{B_0}$ and remarks that this "assumes only that A_{B_0} is well-defined" (p. 193). Proposition 4 then gives a threshold for the last proposition added in a sequence. Sequence-Irrelevance says that "rational constraints on beliefs are not affected by the sequence (if any) in which they are formed" (p. 194), and Proposition 5 concludes that there are "belief-independent values T_p and F_p" fixing a threshold for each proposition (p. 195).

Two features of Proposition 5's scope matter below. Its T_p and F_p are values of the measure in a context, not functions of the credences. The proof obtains them by setting T_p equal to $T_p^{B^-}$ for a state B^- (p. 207). And Dorst's reply to the worry that variable Lockeanism is empty depends on it: the thresholds "come from the (dis)values of true (false) beliefs" (p. 196). So a score under which, with values and context fixed, a proposition is believed at one credence and not at a higher one refutes the proposition. Second, I read Sequence-Irrelevance as Dorst states it, as a constraint on rationality: the order in which a final state was formed does not change its rational status. Section 3 shows that the counterexample respects it on that reading.

3. The counterexample

Define the record measure with premium $\beta > 0$:

$$A_\beta(B, w) = \sum_{p \in B} v_p(w) + \beta \cdot [\, w \in \bigcap B \,],$$

where $\bigcap \emptyset = W$ and $[\cdot]$ is 1 if the condition holds and 0 otherwise. Call $\bigcap B$ the core of B: the strongest proposition the state commits its holder to. The premium rewards the truth of that total commitment. A state whose core is true is one from which any multi-premise deduction yields only truths, and a body of belief can be valuable as a whole in this way, which is the holist's thought. Dorst motivates Separability by an exam scored question by question (p. 184); many exams also reward a perfect paper. The premium does not grow with the number of beliefs, because it rewards one proposition, the core; the informativity of a state is rewarded by the separable part. The empty state's core is W, whose truth is trivial, so it earns the premium everywhere, as an uninformative answer should. The measure is purely alethic: $A_\beta(B, w)$ depends only on B and on which of its members are true at w.

It meets the unrestricted truth-directedness condition motivating §4.2, but not the stronger marginal-value restriction introduced there. Adding a true p raises A_β by exactly T_p, since if every member of B_0 was true at w, all still are. Adding a false p lowers it by $|F_p|$, or by $|F_p| + \beta$ if B_0 was error-free at w. So the measure is strictly truth-directed, and every increment is conservative when $T_p < -F_p$. It does not depend on the current state, so Maximize Expected Accuracy issues one verdict about final states however they are reached, and every belief state is an option.

The case. Let $W = \{w_1, w_2, w_3\}$, $T_p = 1$ and $F_p = -2$ for every p (so $\tau = 2/3$), and $\beta = 3$. Write $\mathbf{1} = \{w_1\}$, $12 = \{w_1, w_2\}$ and $13 = \{w_1, w_3\}$. At $C = (0.70, 0.15, 0.15)$ the unique optimum among all 2^8 states is $\{\mathbf{1}, 12, 13, W\}$, with expected accuracy 4.30. The runner-up, $\{12, 13, W\}$ at 4.20, has the same core and lacks only $\mathbf{1}$, so the margin is $e_{\mathbf{1}} = 0.10$: here the separable term decides. At $C' = (0.75, 0.01, 0.24)$ the unique optimum is $\{13, W\}$, with 4.94; the runner-up is $\{\mathbf{1}, 12, 13, W\}$ at 4.75. Here the premium decides: the core 13 is nearly certain, and its premium outweighs what the more specific core adds. (The optimum at C' also omits 12, whose credence 0.76 exceeds τ.)[2] So $\mathbf{1}$ is believed at credence 0.70 and not at 0.75. By Non-vacuity and Maximize Expected Accuracy, each unique optimum is rationally permitted, and Proposition 5 would require $\tau_{\mathbf{1}} \leq 0.70$ and $\tau_{\mathbf{1}} \geq 0.75$.

The case is not a knife-edge.

[2] All numerical claims were checked by exhaustive enumeration of belief states in exact rational arithmetic (all 2^8 states on three worlds, all 2^{16} on four) and, on larger spaces, by maximizing Φ over cores as Theorem 3 licenses.

Theorem 1. Let $0 < T < -F$, $\beta > 0$, $D = T - F$, $\tau = -F/D$, and

$$c_0 = \frac{\beta - T - 2F}{2D + \beta}, \qquad c_1 = \frac{\beta - 2F}{2D + \beta}.$$

Then $\max(c_0, \tau) < c_1 < 1$, and for all $c < c'$ in $(\max(c_0, \tau), c_1)$, $\mathbf{1}$ belongs to the unique optimum at $(c, \frac{1-c}{2}, \frac{1-c}{2})$ and to no optimum at $(c', \varepsilon, 1 - c' - \varepsilon)$, for all sufficiently small $\varepsilon \geq 0$.

The proof is in the Appendix. For $0 < \beta \leq D$ the interval has length $\beta T/(D(2D + \beta))$: it narrows as the premium shrinks but never closes. This instantiates what Holguín calls Non-Monotonicity: an agent can be permitted to think p at a lower credence and not at a higher one (2022, §5).

Fixed context. A defender may reply that variable Lockeanism already lets belief in p depend on credences in alternatives. In Dorst's model of the question "Who do you think will win?", the question lowers the penalty for falsely believing answers of the form "x wins" "until it gets an answer" (p. 190). Whether one believes that Steve wins at credence 0.48 then depends on whether a rival is more probable. But in that model the dependence enters through a change of values, driven by the demand that the question be answered, which is not itself an expected-accuracy calculation; within each context belief is monotone in credence. The record measure produces the dependence with values, question and context fixed, as a consequence of maximizing expected accuracy. This matters for Holguín's objection that sophisticated Lockeanism requires "brute stipulations about the coordination of the proposition-sensitive thresholds" and "remains powerless to account for non-monotonicity" (2022, §7.2). Here no coordination is stipulated.

Order. The record measure does make one procedure order-dependent. At C, an agent who starts from the empty state and considers $\mathbf{1}$ first gains $e_{\mathbf{1}} - \beta C(\{w_2, w_3\}) = 0.1 - 0.9 = -0.8$, declines it, and ends greedily at $\{12, 13, W\}$. Considering $\mathbf{1}$ last, she gains 0.1 and ends at the optimum. Dorst's example of designing a sequence to make belief in God rational (p. 194) might suggest a violation of Sequence-Irrelevance. It is not one. Proposition 4 is a necessary condition: a final state is rational only if it maximizes expected accuracy relative to its predecessor. A global optimum meets that condition however it is written as the last step of a sequence, so Proposition 4 applies to it in every order. The greedy outcome is not an optimum, so under Maximize Expected Accuracy it is not rational whatever sequence produced it. No sequence confers rationality on any state. What depends on order is a procedure that Maximize Expected Accuracy does not endorse; Dorst himself says the process of forming beliefs in sequence "may be purely imaginative" (p. 194).

4. Diagnosis

The quantifier gap. Truth-directedness gives, for each world where p is true, some positive increment. The passage on p. 193 then names one number $T_p^{B_0}$ as "the value" of adding a true belief in p. That moves from "for every w there is a t" to "there is a t for every w." Under the record measure, adding the false $\mathbf{1}$ to $\{13\}$ costs 2 at w_2 but 5 at w_3, since only at w_3 was the prior state error-free. In expectation,

$$EA_C(B) - EA_C(B \setminus \{p\}) = e_p - \beta\, C(\textstyle\bigcap(B \setminus \{p\}) \setminus p),$$

a threshold on $C(p)$ that moves with the credence of the rest of the core outside p. The step needs:

(ME) Marginal extensionality. For every B_0 and $p \notin B_0$, $A(B_0 \cup \{p\}, w) - A(B_0, w)$ depends on w only through whether $w \in p$.

ME is not a consequence of well-definedness. It says the marginal value of a belief depends on the world only through that belief's own truth value.

The content gap. Grant ME. Proposition 4 then yields thresholds $\tau_p^{B^-}$ indexed to B^-, the rest of the state, and the proof of Proposition 5 generalizes from one B^- to all. That requires $\tau_p^{B^-}$ not to vary with B^-. Sequence-Irrelevance cannot supply this: B^- is not a sequence but the content of the other beliefs, which is what the holist says matters. Dorst himself separates the two: "no holist ever thought that the insight of their position is that the order in which you consider propositions affects what you ought to believe" (p. 195). The natural strengthening is:

(BI) Background-independence. $T_p^{B_0}$ and $F_p^{B_0}$ do not depend on B_0, for states B_0 not containing the complement of p.

Theorem 2. Let A be truth-directed, and consider states containing no proposition together with its complement.[3] (a) A satisfies ME iff $A(B, w) = A(\emptyset, w) + \kappa(B) + \sum_{p \in B} J_p \, [w \in p]$ for a world-independent κ and constants $J_p > 0$ (the sum over contingent $p \in B$). (b) A satisfies ME and BI iff κ is additive, that is, iff $A(B, w) = A(\emptyset, w) + \sum_{p \in B} v_p(w)$ for proposition-dependent values v_p.

The proof (Appendix) turns on one observation: adding p and q in either order gives the same total, and under ME one side of the resulting identity is affine in $[w \in p]$, the other in $[w \in q]$. Since $A(\emptyset, w)$ is common to all states, (b) returns us to the separable measures of Corollary 2.

The proof yields more than the theorem states. Under ME the gain $J_p^B = T_p^B - F_p^B$ is the same for every background B, so $\tau_p^B = -F_p^B / J_p$.

Corollary. Under ME, on the stated domain, the threshold for p is background-independent iff BI holds. Hence the step from $\tau_p^{B^-}$ to a single τ_p in the proof of Proposition 5 is valid iff A is separable on that domain.

So, under ME, reading Sequence-Irrelevance strongly, as requiring background-independent thresholds, is no weaker than BI. And the Corollary yields a dilemma. Under ME, the only holism left is the world-independent κ of Theorem 2(a), so the belief-dependence Dorst explicitly permits must take that form, and an additive κ is just separability. If he admits it, Proposition 5 fails: a small bonus for deductively closed states satisfies ME and truth-directedness and makes belief non-monotone.[4] If he excludes it, his §4.2 measures are separable on the domain, and §4.2 adds nothing to Corollary 2. Without ME, the record measure is a counterexample. Theorem 2 concerns a single measure. Dorst's measures are indexed to the current state; for such a family, ME at each member's own state, together with BI read across the family (the increments $T_p^{B_0}$ and $F_p^{B_0}$ of A_{B_0} at B_0 do not depend on B_0), yields fixed thresholds through Proposition 4 without collapsing the family into one separable measure. The counterexample is unaffected, since a state-independent measure is a constant family.

[3] The proviso is needed: there are ME measures in which the marginal value of p changes when its complement is believed. It is harmless here, since such states are never optimal for separable measures under Dorst's Variable Conservativeness (his Proposition 3), nor, by Theorem 3, for the record measure.

[4] With $T = 1$, $F = -2$ and a bonus of $1/2$ for deductively closed states, the measure satisfies ME and truth-directedness, and on three worlds $\{w_1\}$ is believed at $(0.575, 0.2, 0.225)$ but not at $(0.65, 0, 0.35)$.

5. What holism delivers instead

The record measure's optima have an exact description. For each proposition X let

$$\Phi(X) = \beta\, C(X) + \sum_{p \supseteq X} \max(e_p, 0).$$

Theorem 3. For any finite W and proposition-dependent values, $\max_B EA_C(B) = \max_X \Phi(X)$. A state B is optimal iff its core maximizes Φ, B contains every $p \supseteq \bigcap B$ with $e_p > 0$, and B contains no p with $e_p < 0$. Call the case generic when Φ has a unique maximizer X^* and no $C(p)$ equals a threshold. With uniform values, in the generic case the unique optimum is $B^* = \{p \supseteq X^* : C(p) > \tau\}$, and: (i) B^* is consistent iff $X^* \neq \emptyset$, and it is closed under single-premise entailment; (ii) X^* contains every world more probable than any world it contains; (iii) B^* is deductively closed iff $C(X^*) > \tau$, in which case it consists of exactly the consequences of X^*.

Call this core-relative Lockeanism. The agent selects a working hypothesis X^*, trading its probability against the above-threshold content it entails, and believes the above-threshold propositions it entails. Lockean belief is the special case in which the best core is the intersection of all above-threshold propositions. Closure can fail: on four worlds with $C = (0.62, 0.18, 0.10, 0.10)$ and the values of Section 3, the unique optimum believes $\{w_1, w_2\}$ and $\{w_1, w_3\}$ but not $\{w_1\}$.

Dorst and Mandelkern (2022, §§3–4.1) already derive filtered guesses by maximizing epistemic utility and, following Holguín, connect belief to their consequences. Their Jamesian expected answer-value is $C(X)J^{I_Q(X)}$, where $I_Q(X)$ is the proportion of question cells ruled out and $J \geq 1$. On the singleton question, $J = 125/64$ selects **1** at C and 13 at C': their scores are respectively $35/32$ versus $17/16$, and $75/64$ versus $99/80$. Taking all consequences gives exactly the belief sets in Section 3. This is an instantiation of their model, not their printed numerical example. Neither the reversal nor the filtered-core structure is new.

Theorem 3 identifies a more specific result. It optimizes a score on unrestricted belief states with strict gains for adding truths and strict losses for adding falsehoods. Scoring a state simply by its conjunction's Jamesian value would fail that requirement: adding an entailed proposition leaves the score unchanged. Moreover, Φ depends on the credences of individual consequences, not just the core's probability and informativeness. Its output can omit the core itself; part (iii) identifies when deductive closure holds. Levi and Leitgeb, like the single-guess entailment bridge, accept all consequences of the selected proposition. The contribution is this optimizer characterization in Dorst's setting, not a first utility-theoretic derivation of guess-based belief.

Increasing the premium, with credences and values fixed, cannot lower the credence of the optimal core (add the optimality inequalities at the two premiums), so it favors more probable cores; when world probabilities are distinct, (ii) makes these broader cores, which entail fewer beliefs. When every world has positive credence, $\Phi(W) - \Phi(X)$ grows linearly in β for every proper X, so a large enough premium leaves only W believed. The score therefore requires a defended balance between acquiring truths and protecting the joint truth of what one believes. Calling both aims accuracy does not fix their relative weight.

6. Refinement

The strongest reply to the counterexample is that it depends on how possibilities are individuated. Split the 1% world of C' into two worlds of credence 0.005. On the refined space $\Phi(\mathbf{1}) = 7.25$ exceeds $\Phi(\{w_1, w_3\}) = 6.91$, so $\mathbf{1}$ is believed at 0.75; it is also believed at the correspondingly refined C. The non-monotonicity is gone. In general, with uniform values, splitting a world of credence zero into k worlds multiplies the separable part of Φ for every core excluding it by 2^{k-1}, since each pair of coarse supersets that differ only on that world becomes 2^k refined supersets of equal credence, and it leaves the premium term alone, so by Theorem 3 the core is chosen as if the premium were $\beta/2^{k-1}$, and for fixed β the interval of Theorem 1 shrinks exponentially in k. The premium is paid once while the separable sum ranges over the whole algebra. Verdicts about a fixed question should not depend on how unrelated possibilities are counted, and separable scores meet that demand.

Two repairs are available. First, the culprit is the flat sum. Replace it with the mean over the algebra $\mathcal{A}$: $A(B, w) = |\mathcal{A}|^{-1} \sum_{p \in B} v_p(w) + b\,[w \in \bigcap B]$. On a fixed space this only rescales the premium, so Theorems 1 and 3 apply unchanged. Splitting a credence-zero world now leaves the Φ-value of every core that excludes it, and so every verdict, exactly unchanged. In the case of Section 3 (with $b = 3/8$), splitting w_2 into 2, 4 or 8 equiprobable worlds, at both C and C', leaves both verdicts in place. Second, Dorst's accuracy function already takes "a set of propositions" as an argument (p. 179). Let that set be the algebra generated by the question in context. Splitting worlds that the question does not distinguish then changes nothing.

Neither repair yields invariance under every refinement. A theorem explains why that stronger requirement excludes the present non-Lockean behavior. A coarsening is a surjection $\pi : W \to W^*$; it carries a credence C on W to $C^* = C \circ \pi^{-1}$ and identifies $p^* \subseteq W^*$ with $\pi^{-1}(p^*)$. Suppose the context assigns a measure to each finite space.

(RI) Refinement invariance. For every coarsening π, credence C and $p^* \subseteq W^*$, $\pi^{-1}(p^*)$ belongs to every (some) optimum at C iff p^* belongs to every (some) optimum at C^*.

(CI) Conservative increments. On every two-world space, adding a contingent p to any state gains less at the world where p is true than it loses at the world where p is false.

Theorem 4. Suppose every measure is truth-directed and CI and RI hold. Then for each contingent proposition p there is a $\tau_p \in (1/2, 1)$, fixed by the measure on the two-world coarsening $\{p, W \setminus p\}$, such that on every space and at every credence, p belongs to every optimum if $C(p) > \tau_p$ and to none if $C(p) < \tau_p$.

The proof (Appendix) reduces every verdict on p to the two-world problem that asks only whether p. Truth-directedness and CI then give a single threshold above $1/2$. Separable measures whose values are preserved under coarsening satisfy RI. The record measure does not: on the two-world coarsening its threshold for $\mathbf{1}$ is $(\beta - F)/(D + \beta) = 5/6$, although on three worlds it believes $\mathbf{1}$ at 0.70. CI is needed for the theorem's conservativeness conclusion: separable values $T_p = 2$, $F_p = -1$ satisfy RI but give threshold $1/3$.

RI is one sufficient repair, not a necessary premise of every Lockean derivation. For example, assign uniform separable values $T = 1$, $F = -n$ to each n-world space, $n \geq 2$. Every fixed space is Lockean, with threshold $n/(n+1)$, but the family violates RI. Dorst's fixed-context conclusion does not itself require those thresholds to survive changes of space. The theorem establishes the stronger cross-space result; it must not be used to equate fixed-space Lockeanism with refinement invariance.

The refinement objection therefore splits in two. Its restricted form, invariance under refinements that leave the question fixed, is met by the question-relative record measure. Its

unrestricted form requires the verdict on whether horse 1 wins to survive replacing "Which horse wins?" with "Does horse 1 win?", when the family of measures treats that change as a coarsening within one context. That is a substantive restriction on question dependence. Dorst's proposition-specific values can accommodate contextual changes; Holguín instead allows the salient alternatives to affect rational thinking directly (2022, §§7.2, 8.1). An advocate of RI must explain why these changes should preserve the relevant context. Its sufficiency is informative, but accuracy maximization and order independence do not supply that explanation. Whether a weaker general invariance condition already forces thresholds remains open.

7. Objections

The premium is not accuracy; it double-counts the conjunction. The premium is alethic: it depends only on which believed propositions are true. It is not a disguised belief in the core, since in the four-world case it accrues on $\{w_1\}$, which is not believed. And it respects strict truth-directedness, the constraint Dorst places on holistic measures. The objector may insist that genuine accuracy is local, so that a belief's contribution depends only on its own truth. That is ME. With BI it yields separability, but ME alone leaves the term κ of Theorem 2(a), and Section 4's dilemma applies. Excluding the premium on these grounds denies the holism §4.2 set out to accommodate. Someone may still find the premium unmotivated. Then the dispute is about the value of an error-free body of belief, which is my claim.

Maximize Expected Accuracy gives only a necessary condition; perhaps neither optimum is rational. A purely necessary norm can indeed permit nothing, and then no counterexample establishes a rational state violating its conclusion. Non-vacuity excludes that response without declaring every optimum rational. At the two credences of Section 3 the optimum is unique, so it must be the permitted state. In Theorem 1 all optima agree on the relevant verdict. The objection therefore survives only by rejecting this existence requirement or by imposing an additional restriction that excludes the score.

The result is negative only; Dorst's picture survives. In part, yes. Belief remains a function of credence and contextual values, and Dorst's gloss of beliefs as "best guesses" (p. 196) fits core-relative Lockeanism at least as well as Lockeanism. What does not survive is the claim that accuracy forces the Lockean form. That matters to Holguín's argument, which targets that form: non-monotonicity tells against sophisticated Lockeanism, not against accuracy-first epistemology.

8. Conclusion

On its unrestricted reading, Dorst's holistic extension does not derive fixed Lockean thresholds. A record premium satisfies strict truth-directedness and order-independent maximization while producing non-monotone belief; Non-vacuity connects that optimization result to rational permission. On a stronger reading, the derivation needs restrictions on marginal values. For a single measure, marginal extensionality plus background-independent marginal thresholds recovers separability on the stated domain. Cross-space refinement invariance is a different sufficient route, with a stronger conclusion and its own substantive commitments. Earlier theories already make belief guess-like; the record measure shows how a particular version arises under the strict accuracy conditions at issue here. Defending Lockeanism from accuracy therefore requires defending the restrictions that select its form, rather than treating order independence as sufficient.

Appendix: Proofs

Proof of Theorem 1. Write $e_p = C(p)D + F$; note $\tau > 1/2$. At $C_c = (c, \frac{1-c}{2}, \frac{1-c}{2})$ with $c > \tau$, the positive terms are $e_{\mathbf{1}}$, $e_{12} = e_{13}$ and $e_W = T$; all others are negative (their credences are at most $1-c < 1/2$). Then $\Phi(\mathbf{1}) - \Phi(W) = e_{\mathbf{1}} + 2e_{12} - \beta(1-c) = c(2D+\beta) + T + 2F - \beta$, positive iff $c > c_0$. Also $\Phi(\mathbf{1}) - \Phi(13) = e_{\mathbf{1}} + e_{12} - \beta\frac{1-c}{2} = \frac{1}{2}(\Phi(\mathbf{1}) - \Phi(W)) + \frac{1}{2}e_{\mathbf{1}} > 0$. Next, $\Phi(\mathbf{1}) - \Phi(\{w_2\}) = e_{\mathbf{1}} + e_{12} + \beta\frac{3c-1}{2} > 0$ and $\Phi(\mathbf{1}) - \Phi(\{w_2, w_3\}) = e_{\mathbf{1}} + 2e_{12} + \beta(2c-1) > 0$; the cores 12 and $\{w_3\}$ are symmetric; and $\Phi(\mathbf{1}) - \Phi(\emptyset) = \beta c > 0$. By Theorem 3 the unique optimum has core $\mathbf{1}$ and contains $\mathbf{1}$. At $(c', 0, 1-c')$ with $\tau < c'$: $e_{\mathbf{1}} = e_{12} = c'D + F > 0$ and $e_{13} = e_W = T$, so $\Phi(13) - \Phi(\mathbf{1}) = \beta(1-c') - 2e_{\mathbf{1}}$, positive iff $c' < c_1$, while $\Phi(\emptyset) = \Phi(\mathbf{1}) - \beta c'$. A state containing $\mathbf{1}$ has core $\mathbf{1}$ or $\emptyset$, so no optimum contains $\mathbf{1}$; these strict inequalities persist for small ε. Finally, $c_1 - c_0 = T/(2D+\beta)$, $c_1 - \tau = \beta T/(D(2D+\beta))$ and $1 - c_1 = 2T/(2D+\beta)$ are positive. □

Proof of Theorem 2. Write $\Delta_p^B(w) = A(B \cup \{p\}, w) - A(B, w)$ for $p \notin B$. Under ME, $\Delta_p^B = F_p^B + J_p^B[w \in p]$ for contingent p, and Δ_p^B is constant for $p \in \{\emptyset, W\}$; truth-directedness gives $F_p^B < 0 < F_p^B + J_p^B$. (i) Let $p, q \notin B$ be distinct, q not the complement of p. Computing $A(B \cup \{p,q\}, \cdot) - A(B, \cdot)$ in both orders gives $\Delta_p^B - \Delta_p^{B\cup\{q\}} = \Delta_q^B - \Delta_q^{B\cup\{p\}}$. The left side is $\alpha + (J_p^B - J_p^{B\cup\{q\}})[w \in p]$, the right side $\gamma + \delta[w \in q]$, or constant if q is trivial. If p, q are contingent, distinct and not complementary, $1, [w \in p], [w \in q]$ are linearly independent, so $J_p^{B\cup\{q\}} = J_p^B$; if q is trivial, the same follows. (ii) Building any B that contains neither p nor its complement from $\emptyset$ one member at a time, (i) gives $J_p^B = J_p^\emptyset =: J_p$. (iii) For B without complementary pairs, add its members in any order: $A(B, w) - A(\emptyset, w) = \kappa(B) + \sum_{p\in B} J_p[w \in p]$, where $\kappa(B)$, the sum of the F-terms and constant increments, is world-independent, and order-independent because the total is. The converse of (a) is immediate. For (b), BI says $F_p^B = \kappa(B \cup \{p\}) - \kappa(B)$ is independent of B, which holds iff $\kappa(B) = \kappa(\emptyset) + \sum_{p\in B} f_p$; then $v_p = f_p + J_p[w \in p]$. Separable measures satisfy ME and BI trivially. The Corollary follows from (ii): $\tau_p^B = -F_p^B/J_p$. □

Proof of Theorem 3. $EA_C(B) = \sum_{p\in B} e_p + \beta C(\bigcap B)$. Every member of B includes $\bigcap B$, so $EA_C(B) \le \Phi(\bigcap B)$, with equality iff B contains every $p \supseteq \bigcap B$ with $e_p > 0$ and no p with $e_p < 0$. Conversely, given X, let $B_X = \{p \supseteq X : e_p > 0\}$ and $X' = \bigcap B_X \supseteq X$. The members of B_X are exactly the positive-e propositions including X', so $EA_C(B_X) = \Phi(X') \ge \Phi(X)$, because $C(X') \ge C(X)$. Hence $\max EA = \max \Phi$, and the optimality condition follows. For uniform values, $p \subseteq q$ implies $C(q) \ge C(p)$, which gives upward closure. For (ii): if $u \in X^*$, $v \notin X^*$ and $C(v) > C(u)$, swapping u for v maps the supersets of X^* bijectively onto those of the new core without lowering any credence, and raises the core's credence, contradicting maximality. For (iii): B^* is upward closed with $\bigcap B^* = X^*$, so it is closed under conjunction iff $X^* \in B^*$, iff $C(X^*) > \tau$; it is consistent iff $X^* \neq \emptyset$. □

Proof of Theorem 4. Let π send p to u and $W \setminus p$ to v. By RI the verdict on p at C equals the verdict on $\{u\}$ in the two-world problem at credence $c = C(p)$. There, truth-directedness excludes every state containing $\emptyset$ (dropping it raises the score at both worlds) and every state lacking $\{u, v\}$ (adding it raises the score at both), leaving $\{W\}$, $\{W, p\}$, $\{W, \bar{p}\}$ and $\{W, p, \bar{p}\}$. If $\{W, p, \bar{p}\}$ were optimal at c, dropping $\bar{p}$ or p could not help: by CI, $(1-c)g \ge c\,l$ with $g < l$, so $c < 1/2$, and $c\,g' \ge (1-c)\,l'$ with $g' < l'$, so $c > 1/2$. Let t and f be the gain and loss from adding p to $\{W\}$. Then $EA(\{W, p\}) - EA(\{W\}) = ct - (1-c)f$, positive

iff $c > \tau_p := f/(t+f)$, and $\tau_p > 1/2$ by CI; similarly $\{W, \bar{p}\}$ beats $\{W\}$ only if $c < 1/2$. So for $c > \tau_p$ the unique optimum is $\{W, p\}$, and for $c < \tau_p$ no optimum contains p. □

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When Acting on Knowledge Makes Free Evidence Harmful: Good's Theorem in Normality Models

Abstract. Knowledge-based decision theory maximizes expected utility with probabilities conditioned on what one knows. I test it against Good's theorem inside Goodman and Salow's normality theory of inductive knowledge. In finite statist, comparable models with K-normality, a free question has non-negative value in every decision problem if and only if no state compatible with one's knowledge is one where the answer would reopen a possibility one now rules out. Given suitable questions, every question is safe if and only if knowledge is gapped, and in finite graded models no factive state that contains rational belief is safe unless it is the whole evidence. In Goodman and Salow's model of repeated weighing, a free second weighing has negative value at ten-to-one odds. I then examine three expected-utility developments of Salow's Conditional Knowledge Dominance. Conditioning on the options' differing fails wherever the unconditional rule fails; conditioning on the evidence's changing one's choice also fails. Conditioning answer by answer is safe, but it makes the agent pay to check what she knows, which Conditional Knowledge Dominance forbids.

1. Introduction

If you know p, any evidence against p is misleading, so it seems you may avoid it. This is the dogmatism puzzle. Salow (2025) offers the most economical recent dissolution. If Preservation holds (discovering something consistent with your beliefs never requires giving one up), you cannot believe both p and that you might meet evidence defeating p, so the puzzle never arises. If Preservation fails, the puzzle needs Weak Knowledge Dominance, which Salow replaces with a principle that looks equivalent but is not:

> Conditional Knowledge Dominance: If you know that B's consequences would be better than A's if they're not equally good, you should choose B over A. (Salow 2025, 33)

In Salow's Suitcase, you do not know that the suitcase weighs under 18 kg on the hypothesis that it weighs over 17.5 kg, since discovering the latter would defeat that knowledge. So the principle does not tell you to avoid checking. Answering an objection about accuracy, Salow grants that his response "by itself doesn't predict that openness to further evidence is always required", but holds that "a natural development of it will" (2025, 35). The development he has in mind solves "a general problem" about how all-out attitudes guide choices under uncertainty, and "Any solution to that problem will generalize to this one" (2025, 34).

The most developed proposal of that kind is knowledge-based decision theory (KDT): maximize expected utility using probabilities conditioned on what you know. Goldschmidt (2024) proposes axiomatic foundations for knowledge-based decision theories, with a representation theorem for Hawthorne and Stanley's (2008), and Hong (2024) explores the view. I ask whether KDT, and expected-utility versions of Salow's conditional principle, keep Good's (1967) theorem that free evidence never has negative expected value, once knowledge is modeled by the most developed formal theory of inductive knowledge, the normality framework of Goodman and Salow (2023, 2025). My thesis:

> In finite statist, comparable normality models with K-normality and the requisite questions available, conditioning expected utility on knowledge preserves Good's theorem exactly where knowledge is gapped; of three expected-utility versions of Salow's conditional repair, two fail, and the third succeeds only by abandoning the knowledge-based verdict on costly inquiry.

Theorem 1 identifies when a free question is safe under KDT. Theorem 2 extends its necessity result to factive states containing rational belief. Bjorn's repeated weighing supplies a continuous example with ordinary payoff ratios (§3). Section 5 then distinguishes three conditional evaluations of inquiry. Its negative results concern specified expected-utility rules, not a refutation of Salow's qualitative principle or every knowledge-based decision theory.

Salow (2025, 33) already notes that a 'for all you know' dominance principle can support dogmatism even with Preservation. Goodman and Salow (2025, henceforth BRN) show that statism, collapse and comparability validate Preservation, and that Preservation fails in Bjorn. Geanakoplos (1989) gives sufficient conditions for valuable information beyond partitional learning; Dorst (2020) and Das (2023) develop related epistemological questions. Hong (2026) gives two arguments suggesting that whatever the normality framework captures is not action-guiding. The general decision-theoretic objection is therefore not new.

Dorst et al.'s (2021) Deference Done Better supplies an even closer general comparison: Total Trust is equivalent to Value, the requirement that in every decision problem each strategy recommended by the expert's probabilities have expected utility, by one's own probabilities, at least as high as any option (Theorem 2.2). Here the frame's worlds are the states in E, the current probabilities are $\pi = \Pr(\cdot \mid K_{_E}(x))$ and the expert's state-indexed probabilities are $P_{_s} = \Pr(\cdot \mid K_{E\cap\Pi(s)}(s))$. Since the prior has full support, $P_{_s}$ fixes the post-answer ken, so KDT with the fixed tie order is a recommended strategy. Universal safety therefore implies their Weak Value condition, and Value implies universal safety; their Appendix B (§§B.1–B.2) proves Weak Value equivalent to Total Trust and to Value. So a question is safe at x exactly when π totally trusts this frame, and Theorem 1 characterizes, within these normality models, when it does. Their Theorem 4.1 already characterizes Total Trust for any finite frame, by requiring the relevant candidate expert functions to be 'modestly informed' and π to lie in their convex hull. The increment here is to state that boundary in normality-theoretic terms, as relative defeat and gappedness, together with the graded-model consequence and the tests of three conditional inquiry rules.

2. Framework

I follow Goodman and Salow (2023) and BRN. A normality structure has a set S of states, a family $\mathcal{E}$ of possible bodies of evidence, and two relations on situations $\langle s, E\rangle$ (with $s \in E \in \mathcal{E}$): $\succeq$ (at least as normal as), a preorder, and $\gg$ (sufficiently more normal than). $\gg$ implies $\succeq$, is irreflexive, and chains: if $w_1 \succeq w_2 \gg w_3 \succeq w_4$ then $w_1 \gg w_4$. Under statism both relations are evidence-independent, so I treat them as relations on states. Comparability says any two states with the same evidence are $\succeq$-related. Collapse says every strict difference in normality is sufficient. For the theorems S is finite.

Rational belief, given evidence E, rules out exactly the evidential possibilities that something else in E is sufficiently more normal than:

$$B(E) = \{s \in E : \text{ no } u \in E \text{ has } u \gg s\}.$$

Knowledge goes beyond evidence by the same mechanism, but its extent depends on where one actually is. I work mainly with K-normality (Goodman and Salow 2023, 104):

$$K_{_E}(x) = B(E) \cup \{v \in E : v \succeq x\}.$$

One knows what is true throughout $K_{_E}(x)$, which I call one's ken. Evidence is transparent, and Goodman and Salow note that "given TRANSPARENCY, K-NORMALITY entails KK" (2023, 107). Their alternatives, I- and KM-normality, validate their principle Margins and

allow failures of KK (see §6.1 below). Discovering p at $\langle x, E\rangle$ yields $\langle x, E \cap p\rangle$. A question is a partition Π of E whose cells are discoverable ($E \cap c \in \mathcal{E}$ for each cell c); $\Pi(s)$ is the cell containing s.

Knowledge-based expected utility. Fix a prior Pr with full support. A decision problem is a finite set A of acts, each a function from states to utilities. At $\langle x, E\rangle$, KDT chooses an act maximizing $\Pr(u_a \mid K_E(x))$, ties broken by a fixed order.

The value of a free question. Let $\alpha_\Pi(s)$ be the act KDT chooses at $\langle s, E \cap \Pi(s)\rangle$. The KDT value of answering Π before choosing is

$$V_x(\Pi, A) = \sum_s \Pr(s \mid K_E(x))\, u(\alpha_\Pi(s), s) \;-\; \max_{a \in A} \Pr(u_a \mid K_E(x)).$$

This is Good's comparison of acting now with looking and then acting, with $\Pr(\cdot \mid K_E(x))$ in place of the unconditioned probability. Call Π safe at x if $V_x(\Pi, A) \geq 0$ for every decision problem A.

Relative defeat at x for Π: some $s \in K_E(x)$ has $K_{E \cap \Pi(s)}(s) \not\subseteq K_E(x)$. For all one knows, answering would leave open something one now rules out.

Gapped: every member of $K_E(x)$ is sufficiently more normal than every member of $E \setminus K_E(x)$.

States settle what an unconsulted source would say. This is already so in Goodman and Salow's model, where what Bjorn knows after one weighing depends on the second scale's reading (2023, 128) and he has inductive knowledge "about the scales' future readings" (2023, 130 n. 40). The conditioning set is therefore fixed at the time of choice and is the same for both options. If states also settled whether one looks, the conditioning set would depend on the act being evaluated. That is a different route to failures of Good's theorem, and I set it aside.

3. Bjorn's second weighing

Goodman and Salow's (2023, §6) model of Bjorn is statist and comparable, and it violates collapse. Its states are triples $\langle w, y_1, y_2\rangle$: Bjorn's weight and the readings of two scales. With $\delta = (w - y_1)^2 + (w - y_2)^2$, one state is at least as normal as another iff its δ is no larger, and sufficiently more normal iff its δ is smaller by more than c^2. Goodman and Salow derive these relations from independent Gaussian errors (2023, 129). I take the standard deviation as the unit and, following BRN (§5, n. 18), set $c \approx 1.96$.

After the first reading r, Bjorn believes his weight lies in $[r - c, r + c]$. Under K-normality, in any situation with $\delta \leq c^2$ this belief is knowledge, and his ken is $\{\delta \leq c^2\}$. After a second reading r' he believes his weight lies within $c/\sqrt{2}$ of the average of the two readings. As Goodman and Salow show for the case where the first reading is accurate and the second within c of it, he "loses knowledge" whenever $|r - r'| > (2 - \sqrt{2})c$ (2023, 127). Since the readings may differ by up to $\sqrt{2}c$ for all he believes, he believes that the second weighing might defeat his knowledge: a failure of Preservation, the case BRN use against it. With independent unit Gaussian errors, losing the old weight-interval belief has unconditioned probability about 0.417 (about 0.356 conditional on his present knowledge). BRN's estimate, about 0.37, concerns the narrower event of losing it while the discovery remains consistent with the old beliefs: $(2 - \sqrt{2})c < |r - r'| \leq \sqrt{2}c$ (§5, n. 18). Over the full state space, including the readings, Preservation fails at every nonzero difference up to $\sqrt{2}c$, with probability about 0.95. These are different events.

Ten-to-one bet. After the first weighing Bjorn is offered a bet: win 1 if his weight is within c of r, lose 10 otherwise. He knows the bet wins, so KDT takes it. A free second weighing is then offered. After it he would decline the bet whenever the readings differ by more than about $1.12c$, because his new ken gives the losing weights enough probability. Conditional on his present knowledge, this has probability about 0.05, and in each such state he forgoes a payoff he now knows he would get. So KDT values the free weighing at about -0.05; at twenty-to-one, -0.12; at a hundred-to-one, -0.27. A fallibilist who uses the unconditioned probabilities also takes the ten-to-one bet (it wins with probability 0.95; expected value $0.95 - 10(0.05) = 0.45$), and by Good's theorem the second weighing has positive value for her. The magnitudes depend on c. On Goodman and Salow's probabilistic account of "sufficiently more normal" (2023, 121), applied to these errors, $c^2 = -2\ln(1-t)$ for probability threshold t; so $c \approx 1.96$ corresponds to $t \approx 0.85$. At $t = 0.95$ ($c \approx 2.45$) the ten-to-one value is about -0.01 and the hundred-to-one value about -0.15.

Three claims about other situations should be kept apart. (i) At every situation after the first weighing, some state compatible with Bjorn's knowledge has a second reading that reopens a state he now rules out, and a bet on the proposition he knows, paying 1 if it holds and losing a finite amount otherwise, makes the second weighing harmful (appendix, A.5). That proposition concerns the readings as well as his weight. (ii) A reopened weight exists only if $\max(c^2, \delta(x)) < (1+\sqrt{2})^2 c^2$; beyond that, which has probability about 10^{-5}, no bet of this form on his weight alone does the job (A.5). (iii) Theorem 2 and its corollary are finite results using the questions defined in §4; the second weighing is not one of them.

4. Where Good's theorem holds

Write M for any assignment of a set $M_F(s) \subseteq F$ to situations, used in place of K in the definitions of §2.

Lemma 1 (defeat is costly). If some $s_0 \in M_E(x)$ has $M_{E \cap \Pi(s_0)}(s_0) \not\subseteq M_E(x)$, then some two-act decision problem gives Π negative value.

Proof. Take a bet β paying 1 on $M_E(x)$ and $-L$ elsewhere, and declining, which pays 0. β has conditional expectation 1, so it is chosen. At s_0 the post-answer state contains a state outside $M_E(x)$, of positive probability, so for large L the agent declines and gets 0 where β paid 1. Nowhere in $M_E(x)$ does she get more than 1. So $V \leq -\Pr(s_0 \mid M_E(x)) < 0$. □

Lemma 2 (laminar sufficiency). Suppose the post-answer states $\{M'(s) : s \in M_E(x)\}$ all lie within $M_E(x)$ and are reflexive ($s \in M'(s)$), transitive ($t \in M'(s)$ implies $M'(t) \subseteq M'(s)$) and nested (any two are disjoint or one contains the other). Then $V \geq 0$ for every decision problem.

This applies Geanakoplos's value-of-information result (1989, Theorem 1); a short proof is in the appendix.

Lemma 3. Under statism and comparability, for each F the kens $\{K_F(s) : s \in F\}$ are reflexive, transitive and nested.

Theorem 1. Assume statism, comparability and K-normality. (a) A question is safe at x if and only if there is no relative defeat at x for it. (b) If $K_E(x)$ is gapped, every question is safe at x. If it is not, then for some $v \in E \setminus K_E(x)$ the question whether p_v, where $p_v = \{u \in E : \text{not } u \gg v\}$, is unsafe at x if it is available.

Part (a) combines Lemmas 1–3. Part (b) needs only one question at a time, and it rests on a lemma that uses no K-normality.

Lemma 4 (non-gaps are defeasible). Assume statism. Let $m \in M_E(x)$ and $v \in E \setminus M_E(x)$, with not $m \gg v$. Then $m, v \in p_v$ and $v \in B(E \cap p_v)$.

Proof. $m \in p_v$ by hypothesis and $v \in p_v$ by irreflexivity. Nothing in p_v is sufficiently more normal than v. □

Learning p_v is learning that things are not far more normal than v: that one's situation is somewhat abnormal. Goodman and Salow's own example of such a discovery, that an experiment did not end on its first trial, defeats knowledge only in a model without statism (2023, 136). Probabilistic models predict widespread failures of statism, and theirs obey it only for the bodies of evidence they consider (2023, 138 n. 47); so the statism assumption must cover the extended family.

Say that M leaves open what rational belief leaves open if always $B(F) \subseteq M_F(s)$. Knowledge on each of Goodman and Salow's three completions does, and so do justified true belief and iterated knowledge.

Theorem 2 (only gapped states can be conditioned on). Assume statism, with belief defined from normality as above, and suppose M leaves open what rational belief leaves open. If $M_E(x)$ is not gapped, then for some v the question whether p_v, if available, is unsafe at x for expected utility conditioned on M. A gapped set containing x contains $C(x)$: the states reachable from x by steps, each to a state that is not sufficiently less normal than the last.

The boundary in Theorem 1 is not the boundary of Preservation. Take three states a, b, c, in decreasing order of normality, with $a \gg b$ and $a \gg c$ but not $b \gg c$. Preservation holds here for every body of evidence and every discovery. At b one knows $\{a, b\}$; learning that a is false, which is true at b, reopens c. This is the expected-utility counterpart of Salow's observation about the 'for all you know' variant, located exactly.

Call E graded if $C(x) = E$ for every $x \in E$. Bjorn's evidence after one reading is graded, since a chain of small increases in δ connects any two states.

Corollary. (i) Under statism, collapse and comparability, every K-normality ken is gapped, so KDT makes every question safe. (ii) In a finite graded model, if the questions whether p_v are available, expected utility conditioned on a factive state that leaves open what rational belief leaves open makes them all safe at x only if that state is all of E.

Part (i) is the decision-theoretic counterpart of BRN's result that statism, collapse and comparability validate Preservation. Part (ii) says that, given these questions, the only belief-containing state one can condition on in graded models without some free question becoming harmful is one's evidence. Conditioning on evidence is the response on which "decisions should be governed by underlying probabilities rather than our knowledge" (Salow 2025, 23). Part (ii) depends on the available questions. It does not establish the same necessity result for the second weighing alone.

5. The conditional repair

5.1 Three developments

Conditional Knowledge Dominance evaluates options on the hypothesis that they differ. Salow's moral is that "you should always think that, if the evidence you would discover tells strongly against your view, there is a good chance you're wrong" (2025, 36). He gives no expected-utility version. Write $K_E^D(x)$ for what one knows on the hypothesis D. Its K-normality form restricts evidential accessibility to D, as BRN's definition of conditional belief does (App. D.3):

$$K_E^D(x) = \{v \in E \cap D : \text{no } u \in E \cap D \text{ has } u \gg v\} \cup \{v \in E \cap D : v \succeq x\}.$$

Under statism, when $x \in D$, this is the ken one would have on discovering D; in every case it contains the belief one would then have. So it respects Salow's constraint that one is "entitled to believe q on the hypothesis that p only if discovering p would not require you to stop believing q" (2025, 33). Write $u_\ell(s) = u(\alpha_\Pi(s), s)$ for the payoff of looking, and compare looking with an act a in three ways:

(Difference) CEU: look if $\Pr(u_\ell - u_a \mid K_E^D(x)) \geq 0$, where $D = \{s : u_\ell(s) \neq u_a(s)\}$.

(Verdict) VER: look if $\sum_b \Pr(D_b \mid K_E(x)) \Pr(u_b - u_a \mid K_E^{D_b}(x)) \geq 0$, where $D_b = \{s : \alpha_\Pi(s) = b\}$; one conditions on what the evidence would lead one to do.

(Answer) ANS: look if $\sum_c \Pr(c \mid K_E(x)) \Pr(u_\ell - u_a \mid K_E^c(x)) \geq 0$, summing over the cells of Π. These formulas evaluate the future KDT policy α_Π; they do not define fully recursive VER or ANS decision theories.

All three handle Bjorn's bet as Salow's diagnosis predicts. Looking and betting now differ exactly where the second reading would make Bjorn decline, and on that hypothesis his conditional ken includes losing weights. With the actual readings perfectly accurate, at ten-to-one the bet's expected value on that hypothesis is -0.60, so CEU and VER prefer to look; ANS does too, since each decline cell's conditional ken is the post-answer ken.

5.2 Difference and verdict

CEU's hypothesis carries information only because equal-payoff states are excluded from it. A perturbation removes them: if looking and acting now differ at every state, $D = E$, conditional knowledge is ordinary knowledge, and CEU coincides with KDT.

Theorem 3. Assume statism, comparability and K-normality. If there is relative defeat at x for Π, some three-act decision problem makes CEU strictly prefer acting now to answering Π, even if CEU also governs every later choice. KDT gives Π negative value too. These verdicts persist on an open neighborhood of the constructed payoffs.

Construction. Let $M = K_E(x)$ and choose $s_0 \in M$ witnessing relative defeat. Take the bet β of Lemma 1, declining, and $\beta_1 = \beta + \varepsilon\zeta$, where $\zeta(s_0) = N$ and $\zeta = -1$ elsewhere, N gives ζ positive current mean, and $0 < \varepsilon < 1$. Choose L so large that every post-answer ken meeting $E \setminus M$ strictly prefers declining; there are finitely many. Every other post-answer ken excludes s_0, since otherwise transitivity would put s_0's reopened possibilities in it; there β strictly beats β_1 and declining. So looking never selects β_1, which is uniquely best now. All acts differ pairwise at every state, and β_1 differs everywhere from the looking policy; so every comparison CEU makes, now or later, uses ordinary knowledge, CEU agrees with KDT, and no cycle can arise. The gain from acting now at s_0 outweighs the small losses elsewhere (A.6).

For a numerical illustration, take three states $a \succ b \succ c$ with only $a \gg c$, a uniform prior, and $x = a$, whose ken is $\{a, b\}$. Ask whether a; on the other answer the ken is $\{b, c\}$. Let the acts pay $(1, 1, -10)$, $(0, 0, 0)$ and $(0.9, 1.5, -10.1)$. Both rules choose the third now (expectation 1.2). After looking they choose the first at a and decline at b and c, a policy with current expectation 0.5. CEU prefers acting now by 0.7.

Theorem 3 refutes CEU, not Conditional Knowledge Dominance. In the constructed problems the qualitative principle is silent: the most normal state m^* of E lies in the ken, its post-answer ken lies inside the ken and excludes s_0, and there looking yields β, which beats β_1. So one does not know that β_1 is better whenever they differ. More generally, in these models, when looking and acting now differ at every state, Conditional Knowledge Dominance never forbids a free look (A.7). CEU extends the principle into its silence, and there it behaves as KDT does.

VER conditions on a hypothesis in the spirit of Salow's moral: that the evidence would change what one does. It escapes Theorem 3 for large L, because the conditional ken on 'I would decline' then contains an excluded state, where β_1 loses heavily. It fails elsewhere. Take states a, s_2, s_1, v_2, v_1 with normality ranks $0, 0.2, 0.5, 1.15, 1.3$ (lower is more normal), $u \gg w$ iff w's rank exceeds u's by more than 1, prior $(0.67, 0.01, 0.30, 0.01, 0.01)$, $x = a$, and the question $\{\{a\}, \{s_1, v_1\}, \{s_2, v_2\}\}$. The ken is $\{a, s_1, s_2\}$; both non-trivial answers reopen something. Use Theorem 3's acts with $s_0 = s_1$, $N = 10$, $\varepsilon = 0.001$, $L = 30.4$. Looking yields β at a and declining elsewhere. On the hypothesis 'I would decline', the conditional ken is $\{s_1, s_2, v_2\}$: $s_2 \gg v_1$ excludes v_1, but v_2 survives with too little weight to sink β_1, whose expected value there is 0.028. VER prefers acting now by 0.008. Every pair of primitive acts differs everywhere, so later choices are KDT's. VER, and CEU too, can even disvalue a question where there is no relative defeat. Take $a \succ x \succ b \succ c$ with $a \gg x, b, c$ and $x \gg c$ only, a uniform prior, the situation x (ken $\{a, x\}$), the question $\{\{x\}, \{b, c\}, \{a\}\}$, and two acts paying $(-2, 3.1, -2, 2.1)$ and $(3, 3, 3, -3)$ at (a, x, b, c). No answer reopens anything, and looking does at least as well as acting now throughout the ken, so KDT values the question at 0.05. On the hypothesis 'I would choose the first act', however, the conditional ken is $\{x, b\}$, and b lies only in an answer she knows to be false; VER and CEU prefer acting now. Conditioning on what the evidence would lead one to do imports possibilities the evidence itself would not make live.

5.3 Answer by answer

Proposition 4. Assume statism, comparability and K-normality. ANS never prefers acting now to a free question.

Proof. Fix a cell c. If $s \in K_E^c(x)$, then $K_{E \cap c}(s) \subseteq K_E^c(x)$: if $s \in B(E \cap c)$ its ken is $B(E \cap c)$ (by the chaining step in A.2), and if $s \succeq x$ then $\{v \succeq s\} \subseteq \{v \succeq x\}$. These kens are reflexive, transitive and nested (Lemma 3), so Lemma 2 with $K_E^c(x)$ in place of $M_E(x)$ makes each summand non-negative. □

The price appears when evidence is not free.

Proposition 5. Under the same assumptions, if there is relative defeat at x for Π, there are a bet β and a price $\kappa > 0$ such that ANS prefers paying κ and answering Π to taking β now, although the two options differ at every state and β pays more at every state of the ken. Conditional Knowledge Dominance therefore requires taking β.

Proof sketch (A.7). Use Lemma 1's bet. The conditional ken on the cell of s_0 contains an excluded state, so for large L that cell's summand is large, while the others are non-negative. Choose also $0 < \kappa < L$. Inside the ken, looking at a price pays at most $1 - \kappa < 1$; outside it pays $-\kappa \neq -L$. □

In Bjorn at normal situations ($c \approx 1.96$), ANS pays up to 0.019 for the second weighing at ten-to-one. At a hundred-to-one it pays up to 1.23, more than the bet pays in any state she has not ruled out, for evidence about a proposition she knows. ANS secures Good's theorem by valuing inquiry with the mixture $\sum_c \Pr(c \mid K_E(x)) \Pr(\cdot \mid K_E^c(x))$, which gives excluded possibilities positive weight; for inquiry, knowledge no longer settles the probabilities. Restricting ANS to free questions, with costly inquiry left to KDT or Conditional Knowledge Dominance, avoids Proposition 5 at another cost: in Bjorn's case the agent then looks at price zero and refuses at every positive price, however small. Thus ANS, this particular conditional repair, keeps free-inquiry safety at the cost of conflicting with Conditional Knowledge Dominance on costly inquiry.

Outside statist models conditional belief and belief after discovery can diverge (BRN App. D.3), and each rule needs separate assessment there.

6. Objections and replies

6.1 "This is self-trust, not evidence"

The objection says that KDT disvalues questions because the agent distrusts her future self, not the evidence. On the Margins completions, statism, comparability and Margins "together entail NO DEFEAT" (Goodman and Salow 2023, 128). There, by Lemma 4, the reopened possibility lies in the agent's own present ken at m, so the relative defeat in Theorem 2 is a failure of KK at x. Such failures harm the empty question too: by Lemma 1 with the trivial partition, a KDT agent without KK at x pays to decide now rather than later on the same evidence, a cousin of the unmarked-clock case discussed by Ahmed and Salow (2019). For the Margins views I concede the objection. Goodman and Salow report, however, that Goldstein and Hawthorne's (2022) model of multiple observations predicts failures of No Defeat "even given MARGINS" (2023, 130 n. 40), so this escape may be less clean; I have not checked the bet in that model.

For K-normality the objection fails. KK holds, so by Lemmas 2 and 3 the empty question never has negative value (at abnormal situations it can even have positive value). In Bjorn the new reading destroys knowledge despite positive introspection. The structure is the classic dogmatist's: I know the bet wins, the evidence might stop me relying on that, so I avoid it. On the KK-friendly completion, KDT avoids evidence; on the others, it avoids deferring to itself.

6.2 Stakes and encroachment

A ten-to-one payoff ratio is not an absolute measure of stakes: at a fixed ken, positive rescaling of utilities preserves choices and the sign of information value. The finite theorems use large L; Bjorn shows that ten-to-one suffices at $c \approx 1.96$ (§3).

The main reply is pragmatic encroachment. Hawthorne (2004, 181), as Salow reports, suggests that given Good's result the puzzle is handled by the proposal "developed by Fantl and McGrath (2002) and Weatherson (2005)" that encroachment occurs "whenever the difference between maximal and actual probability affects your preferences over available options" (Salow 2025, 24 n. 3). Since the theorems hold the ken fixed across problems, such views escape them. Bjorn's case makes one consequence concrete. Offered the ten-to-one bet alone, he bets both under knowledge-conditioned evaluation (win probability 1) and under evidence-conditioned evaluation (0.95). Offered the bet and a free second scale, the former evaluation strictly prefers acting now, while ordinary Bayesian evaluation strictly prefers looking (§3). An encroachment criterion that withdraws knowledge whenever these two evaluations change the preference therefore withdraws it on the mere offer of free evidence. This consequence is conditional on extending the criterion to these evaluations of inquiry; no general result about every graded state follows from it. Salow objects that it is unclear what would confine encroachment to "these relatively rare counterexamples to Preservation" (2025, 36); in the measurement model the counterexamples are not rare (§3).

6.3 Statism and questions

BRN suggest that statism is "the exception rather than the rule" (§7). Bjorn, however, is statist. Beyond it, the questions are whether post-answer kens stay nested and whether Lemma 4's questions still reopen excluded possibilities. On the probabilistic grounding, normality is relative to a question: Goodman and Salow note that relativizing to how much does Bjorn weigh yields a model like Goldstein and Hawthorne's, in which current knowl-

edge is not sensitive to future observations (2023, 130 n. 40). A weight bet may make that question salient. I have not computed the bet there.

6.4 "Good's theorem is not sacrosanct"

Ahmed and Salow (2019) argue that Good's theorem "may be false", using, among other cases, observations that are not partitional. Bjorn's readings partition the evidential possibilities exactly. The failure arises elsewhere: knowledge is lost, so the post-answer probabilities are not the present knowledge-conditioned probabilities conditioned on the reading, and the later self's policy is not the reading-dependent policy the present self would choose. A defender may accept that foresight of such choices can make looking bad. That concedes predictable, evidence-driven knowledge loss with practical cost, which Salow's dissolution aimed to show harmless. If the agent could commit to ignoring an answer, the present best act stays available and a free signal cannot hurt; the results concern looking followed by the specified later rule, without commitment.

7. Conclusion

In finite statist, comparable models with K-normality and the questions whether p_v available, expected utility conditioned on knowledge makes every free question safe exactly where knowledge is gapped; any factive state that leaves open what rational belief leaves open needs the same condition. On graded normality, which Goodman and Salow defend for measurement, that condition means not going beyond the evidence, and in their own model a free second weighing has negative value at ordinary odds. Of three expected-utility developments of Conditional Knowledge Dominance, conditioning on the options' differing and on the evidence changing one's choice both fail; conditioning answer by answer succeeds, but it pays to check what one knows, contrary to the principle it develops. What has not been shown is that no other way of letting all-out attitudes guide graded choice could deliver the development Salow anticipates, or what happens in non-statist models. The burden on knowledge-conditioned expected utility is precise: weaken the universal safety requirement, restrict the models or questions (collapse is one sufficient restriction), let stakes or offers change knowledge, or stop letting knowledge settle the probabilities used to evaluate inquiry.

Appendix: proofs and computations

A.1 Lemma 2. Let $\mathcal{F} = \{M'(s) : s \in M\}$, $M = M_E(x)$. By nestedness and reflexivity the inclusion-maximal members of $\mathcal{F}$ partition M. For a maximal C, let a_C be the act chosen on C. By induction on $|C|$, $\Pr(u(\alpha(s), s) \mid C) \geq \Pr(u_{a_C} \mid C)$: by transitivity every $s \in C$ has $M'(s) \subseteq C$; states with $M'(s) = C$ choose a_C; the others lie in the maximal proper subkens $D_1, \ldots, D_k$, disjoint members of $\mathcal{F}$ satisfying the hypotheses, on each of which the plan does at least as well as the best act on D_j, hence as a_C. Summing, the plan's expectation on M is at least $\sum_C \Pr(C \mid M) \max_a \Pr(u_a \mid C) \geq \max_a \Pr(u_a \mid M)$. □

A.2 Lemma 3. Reflexivity: $s \succeq s$. Transitivity: let $t \in K_F(s)$. If $t \in B(F)$, every $v \succeq t$ is in $B(F)$ (if $u \gg v$, chaining gives $u \gg t$), so $K_F(t) = B(F) \subseteq K_F(s)$. If $t \succeq s$, $\{v : v \succeq t\} \subseteq \{v : v \succeq s\}$. Nestedness: by comparability the sets $\{v : v \succeq s\}$ are linearly ordered by inclusion, and each ken is their union with $B(F)$. □

A.3 Theorem 1. (a) Kens in different cells are disjoint; by Lemma 3 the post-answer kens are reflexive, transitive and nested; without relative defeat they lie within $K_E(x)$, so Lemma 2

applies. Conversely, Lemma 1. (b) Suppose the ken is gapped, $s \in K_E(x) \cap c$, $v \in c \setminus K_E(x)$. Then $s \gg v$, so $v \notin B(E \cap c)$; and not $v \succeq s$, since $v \succeq s \gg v \succeq v$ would give $v \gg v$. So $K_{E\cap c}(s) \subseteq K_E(x)$. If the ken is not gapped, Lemma 4 gives relative defeat for the question whether p_v, because $B(E \cap p_v) \subseteq K_{E\cap p_v}(m)$. □

A.4 Theorem 2. Lemmas 4 and 1 give the first claim. If M is gapped, $m \in M$ and not $m \gg v$, then $v \in M$; so M is closed under the steps generating $C(x)$. □ Corollary (i). Let $m \in K_E(x)$, $v \notin K_E(x)$. Then not $v \succeq x$, and some $u \gg v$. If $m \succeq x$, comparability and collapse give $x \gg v$, hence $m \gg v$. If $m \in B(E)$, then $v \succeq m$ would give $u \gg m$, so m is strictly more normal than v and collapse gives $m \gg v$. □

A.5 Bjorn. After the first reading, let $\mu = (e_1+e_2)/2$ and $d = e_1-e_2$ for the scale errors, so $\delta = 2\mu^2+d^2/2$; for the numerical model, $\mu \sim N(0, 1/2)$ and $d \sim N(0, 2)$ independently (the swamped-prior treatment of weight). A second reading fixes d, and the post-answer ken of a state s is $|\mu| \leq \max(c/\sqrt{2}, |\mu_s|)$. Let $R = \max(c^2, \delta(x))$, so the current ken is $M = \{\delta \leq R\}$. Choose s with $\mu_s = 0$ and $d_s^2/2$ strictly between $R - c^2$ and R; its post-answer ken reaches $\delta = c^2 + d_s^2/2 > R$. By strictness, a small open neighborhood U of s stays inside M and has post-answer probability of M^c at least some $\eta > 0$, and U has positive probability. A bet paying 1 on M and $-L$ elsewhere, with $L > (1-\eta)/\eta$, is declined throughout U and accepted now, so looking has value at most $-\Pr(U \mid M) < 0$. For weights: $e_1 = \mu + d/2$, and by Cauchy–Schwarz $|e_1| \leq \sqrt{R}$ throughout M. A post-answer ken of $s \in M$ reaches $|e_1|$ up to $\max(c/\sqrt{2}, |\mu_s|) + |d_s|/2$. If $|\mu_s| > c/\sqrt{2}$, this is at most $\sqrt{R}$; otherwise its maximum, $c/\sqrt{2} + \sqrt{R/2}$, exceeds $\sqrt{R}$ iff $R < (1+\sqrt{2})^2 c^2$. Under the Gaussian model $\Pr(\delta \geq (1+\sqrt{2})^2 c^2) \approx 1.4 \times 10^{-5}$. The normal-situation figures in §3 integrate the density over $\delta \leq c^2$ for the bet on $|e_1| \leq c$.

A.6 Theorem 3. Put $P = \Pr(\cdot \mid M)$, choose N with $NP(s_0) > 2$ and fix $0 < \varepsilon < 1$. For each post-answer ken F meeting $E \setminus M$, $q_F = \Pr(E \setminus M \mid F) > 0$; the expectations of β_1 and β on F are at most $1 + N - Lq_F$, so $L > (1+N)/\min_F q_F$ makes declining strictly best on every such F, for every $\varepsilon < 1$. On every other post-answer ken, $\beta_1 = 1 - \varepsilon$ and $\beta = 1$. Now β_1 is uniquely optimal since $P(\zeta) > 0$. At s_0 acting now gains $1 + \varepsilon N$; elsewhere in M looking gains at most ε. So $P(u_{\beta_1} - u_\ell) \geq P(s_0)(1+\varepsilon N) - \varepsilon > 0$. Outside M looking declines while $\beta_1 = -L - \varepsilon$; inside, looking pays 0 or 1, neither equal to β_1. So the difference event is E, and likewise for every pair of primitive acts. Finitely many strict inequalities and nonzero differences persist under small perturbations. □

A.7 CKD silence; Propositions 4–5. Silence. Let m^* be $\succeq$-maximal in E and c^* its cell. If $v \notin K_E(x)$, some $u \gg v$ and chaining gives $m^* \gg v$; so $B(E \cap c^*) \subseteq K_E(x)$, and every state there has post-answer ken $B(E \cap c^*)$ and chooses one act b^*, optimal on that set. If looking differs from a everywhere, b^* beats a at some state of $B(E \cap c^*)$, so one does not know that a is better whenever they differ. Proposition 5. Let $c_0 = \Pi(s_0)$. If $s_0 \succeq x$, $K_E^{c_0}(x) \supseteq K_{E\cap c_0}(s_0)$; if $s_0 \in B(E)$, the part of $K_{E\cap c_0}(s_0)$ above s_0 lies in $B(E)$ by chaining, so its excluded states lie in $B(E \cap c_0) \subseteq K_E^{c_0}(x)$. Either way $K_E^{c_0}(x)$ contains an excluded state, with probability $q > 0$ in it. By the proof of Proposition 4, the c_0 summand is at least $-\Pr(u_\beta \mid K_E^{c_0}(x)) \geq Lq - 1$ and the others are non-negative; so choose $0 < \kappa < \min\{L, \Pr(c_0 \mid K_E(x))(Lq-1)\}$. □

Computations. Supporting scripts enumerate statist comparable structures with up to four states, every nonempty body of evidence, situation and non-trivial partition (64,004 situation–question pairs; 3,306 with relative defeat). Structural enumeration is exhaustive within that domain; sampled priors and utilities test examples, not the universal decision-

problem quantifiers proved above. The examples in §5.2 were also evaluated directly. Bjorn's decision values and knowledge-conditional probabilities use numerical integration. His weight-belief-loss probability is $\operatorname{erfc}((2-\sqrt{2})c/2)$; subtracting $\operatorname{erfc}(c/\sqrt{2})$ gives loss with a consistent discovery.

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A Switching Dilemma for Normality in Actual Causation

Abstract. Normality conditions are a standard way of stopping the Halpern–Pearl definitions of actual causation from counting the flip of a switch as a cause. I prove that one transferable normality policy cannot do this under any of the three definitions. Suppose occurrences are deviant and absences default, and a world's normality depends only on how each variable's status changes, whatever the variable represents. Under the modified definition, the only witness for Suzy's preempting throw has exactly the profile of the witnesses for two standard switches, so every such policy counts all three or none. Under the original and updated definitions, the witness for a switch with both tracks blocked is the witness for genuine prevention plus one more default-restoring change, so every such policy that respects Pareto improvement and keeps prevention counts the flip. Within these codings, a separating Pareto filter must use information beyond the transition profile. I examine one additive repair based on mechanism reliability and prove that, in fixed-context acyclic models, exceptionless equation-conformity leaves only root but-for causes.

1. Introduction

The Halpern–Pearl (HP) definitions count Suzy's throw a cause of the bottle's shattering, although Billy's rock would have shattered it anyway, by evaluating the shattering in a contingency that holds part of the background fixed. The same machinery threatens to make an engineer's flip of a switch a cause of a train's arrival, when the train would have arrived by the other track. The two cases can have isomorphic patterns of counterfactual dependence, and a prominent response distinguishes them by normality: a contingency may witness causation only if the world it produces is at least as normal as the actual world (Halpern and Hitchcock 2015, 20–21). Halpern (2015, 8) handles one model of the switch this way. Normality earns its place if it marks a real difference between switching and preemption, and not just the difference between the verdicts we want.

This essay asks whether one normality policy, applied the same way in every model, can do this. Call a policy anonymous if it ranks a witness world by how each variable's status changes (default or deviant, conforming to its equation or not), whatever the variable represents. My thesis is that, when occurrences are coded deviant and absences default, no anonymous normality filter that respects Pareto improvement keeps genuine prevention and late preemption while excluding every switch, under any of the three HP definitions. The modified definition fails on two switches because its restriction to actual-value contingencies leaves Suzy only a witness whose profile is identical to theirs. The original and updated definitions fail on a switch with both tracks blocked because the contingency that convicts the flip improves on the one that makes an antidote a cause.

The thesis does not say that normality cannot separate switches from preemption. Under the maintained models and coding, a separating Pareto filter must use information beyond the transition profile. Section 5 examines one additive repair: treating Billy's thrown rock failing to hit as more normal than a pressed bell failing to ring, while holding the other scoring contributions fixed. This is not the only possible non-anonymous repair. Graded and event-based variants lie outside the filter theorem.

Closest work and increment. Halpern (2015, 7–8) reports a problem he credits to Schumacher: with both tracks blocked, the original and updated definitions count the flip a cause of non-arrival, and "normality considerations don't help". He notes that the modified definition avoids it. Xiao and Xiong (2026, 4–5, Observation 1) show the same for the Halpern–Hitchcock definition. Halpern (2016, Example 3.2.6, 84–85) observes that if the witness in which Billy throws but does not hit is too abnormal, Suzy's throw ceases to be

a cause under the modified definition and the two throws together become one. Beckers (2026, 2, 10–11) argues for a dilemma of related shape for the regularity theory of Andreas and Günther (2024), discussed in Section 7. The increment here is (i) an exact identity of witness profiles across late preemption and two switches under the modified definition, which covers every anonymous filter at once; (ii) an exact identity linking the both-blocked switch to genuine prevention under the older definitions; (iii) the resulting dilemma across all three definitions; (iv) the assessment of a particular non-anonymous escape; and (v) a lemma showing what exceptionless conformity costs, applied to a conjecture of Halpern and Hitchcock (2015, 37) and to the recent definition of Xiao and Xiong (2026).

Wysocki (2023, §§4–5) gives a stronger obstruction in another setting: conjoined cases can share one witness world while requiring opposing causal verdicts, so no normality ordering of that world can separate them. His cases cover both updated and modified definitions. The present theorem does not improve that general impossibility claim. It compares profiles across standard, uncombined models under an explicit anonymity restriction, and relates the switch-preemption tradeoff across all three HP definitions to a root-transition identity. Its value depends on whether these transferable policies are a useful baseline, not on being the first demonstration that normality can fail.

Why it matters. Schumacher's case is among those Halpern (2015, 8) presents as handled better by the modified definition. The present comparison shows how that improvement trades one difficulty for another under a transferable policy. It identifies where information beyond default status and equation conformity must enter. Such information may be principled; the theorem does not establish that a non-anonymous theory merely encodes desired verdicts. It provides a controlled test for proposed grounds of the distinction.

2. The setting

Models. An acyclic causal model has binary variables. A root has no parents among the modeled variables and takes its actual value from the context; a non-root has an equation in its parents. The actual world is the solution for the actual context; an intervention $[\vec{X} \leftarrow \vec{x}]$ replaces the equations of $\vec{X}$ by constants.

Three definitions. I follow Halpern's (2015, 3–4) statement of the original, updated and modified HP definitions. $\vec{X} = \vec{x}$ is a cause of φ if both actually hold (AC1), no proper subset of $\vec{X}$ satisfies the conditions (AC3), and there is a witness: a setting $\vec{x}'$ and a set $\vec{W}$ of other variables with values $\vec{w}$ such that $[\vec{X} \leftarrow \vec{x}', \vec{W} \leftarrow \vec{w}]\neg\varphi$. The original and updated definitions let $\vec{w}$ be any values, subject to sufficiency conditions (AC2(b), AC2(b^u)). The modified definition requires $\vec{w}$ to be the actual values: it may only freeze variables at what they actually were. A conjunct of a cause is part of a cause. Unless I say otherwise, "$X = x$ is a cause" below means that the single candidate $X = x$ is. The witness world is the solution under the witness intervention.

The filter. Halpern and Hitchcock (2015, 20–21) require a witness world to be at least as normal as the actual world. Their "rough-and-ready guidelines" (20) make a world more normal if some variables take more typical values and none take less typical ones, and leave worlds incomparable when some improve and others worsen, unless special considerations rank them. They also treat it as typical, by default, for endogenous variables to obey their equations (19), and they argue that typicality must sometimes reflect how a variable depends on its parents, not just its value (35).

Categories. These guidelines suggest four categories for a non-root variable in a world, given a default value for each variable:

- **n**: default, conforming (its value is the default, and its equation gives that value);
- **l**: licensed deviant (its value is deviant, and its equation gives that value);
- **f**: frozen default (its value is the default, but its equation demands the deviant value);
- **u**: unlicensed deviant (its value is deviant, but its equation demands the default).

A root is at its default or not. A witness's profile is the multiset of transitions (actual category to witness category) of the variables whose category changes.

Anonymous filters. A filter is anonymous if whether it admits a witness depends only on the witness's profile. It is Pareto if adding to an admitted profile one root that moves from deviant to default yields an admitted profile; this is Halpern and Hitchcock's first guideline applied across models. Two classes illustrate the definition. A componentwise filter fixes a preorder $\succeq$ on $\{\mathbf{n}, \mathbf{l}, \mathbf{f}, \mathbf{u}\}$, counts a root's default as more typical than its deviant value, and admits a witness iff every variable's witness category is $\succeq$ its actual category. An additive filter scores a world as the sum of its variables' scores, $r > 0$ for a root at default and 0 otherwise, and a, b, c, d for a non-root in $\mathbf{n}, \mathbf{l}, \mathbf{f}, \mathbf{u}$; it admits a witness iff the witness world's total is at least the actual total. Unlike componentwise filters, additive ones allow gains in some variables to compensate for losses in others. Both classes are anonymous and Pareto.

Anonymity is a baseline, not a characterization of every principled policy. Its categories use Halpern and Hitchcock's two defaults, typical values and equation conformity. Halpern's compact representations likewise supply default typicalities while allowing exceptions and cross-variable comparisons (2016, 149–150). The theorem tests what the baseline can accomplish without those additional distinctions. A failure identifies information that a richer policy must supply; it does not by itself discredit that information.

Event coding. Occurrences are deviant and absences default, as in Andreas and Günther's (2024, 26) prima facie rule that "occurring events are prima facie more deviant than their absences". Every variable below has default 0.

3. The cases

Roots are listed with their actual values.

Genuine prevention. $P = 1$ (poison in the coffee), $A = 1$ (antidote). $V = P \wedge \neg A$ (the victim dies). The antidote caused the survival.

Chain. $C = 1$; $I = C$, $Y = I$. $C = 1$ caused $Y = 1$.

Boulder (Hitchcock 2001, in the form of Andreas and Günther 2024, 18). $F = 1$ (a boulder falls); $B = F$ (it heads for the hiker), $K = F$ (the hiker ducks), $H = B \wedge \neg K$ (the hiker is hit). The ducking caused the hiker not to be hit.

Short circuit (Halpern and Hitchcock 2015, 34–35). $A = 1$ (a bodyguard adds antidote); $P = A$ (a second bodyguard adds poison only if the antidote is in), $V = P \wedge \neg A$. The antidote is not a cause of the survival.

Late preemption. $ST = 1$, $BT = 1$ (Suzy and Billy throw); $SH = ST$, $BH = BT \wedge \neg SH$ (whose rock hits), $BS = SH \vee BH$ (the bottle shatters). Suzy's throw caused the shattering; Billy's did not.

Tracks (the switch story of Hall 2000, in Hall's 2007 route model as presented by Halpern 2015, 8). $F = 1$ (the engineer flips the switch to the right); $LT = \neg F$, $RT = F$ (the train goes left, right), $A = LT \vee RT$ (it arrives). The flip is not a cause of the arrival.

Doorbells (Gallow 2021, 84–85). $M = 1$ (Marty presses the button), $T = 1$ (Doc has set the switch to the right); $L = M \wedge \neg T$, $R = M \wedge T$ (the left, right bell rings), $E = L \vee R$ (Einstein barks). Gallow's verdict: "Doc's flipping the switch to

the right was (along with Marty's pressing the button) a cause of the right bell's ringing, but it was not a cause of Einstein's barking" (85).

Blocked tracks (Halpern 2015, 7–8; Xiao and Xiong 2026, 4–5). $EF = 1$ (the engineer flips the switch to the right), $LB = 1$, $RB = 1$ (both tracks are blocked); $TA = (EF \wedge \neg RB) \vee (\neg EF \wedge \neg LB)$ (the train arrives). Actually $TA = 0$. The flip is not a cause of the non-arrival.

Gallow models Doorbells with a single many-valued switch variable; the binary route model is mine, in the format of Tracks. Without normality, every definition counts every candidate above, except Billy's throw, a cause or part of a cause, including the three switches and the short-circuit antidote.

4. The dilemma

Theorem 1 (modified definition). Under event coding and the modified definition, Suzy's throw in late preemption, the flip in Tracks and the switch-setting in Doorbells each have exactly one witness world, and the three witness profiles are identical:

$$\{\text{root: deviant} \to \text{default},\ \mathbf{n} \to \mathbf{f},\ \mathbf{l} \to \mathbf{n},\ \mathbf{l} \to \mathbf{n}\}.$$

Hence every anonymous filter admits all three witnesses or none.

Proof. A modified witness may freeze variables only at their actual values. For Suzy, freezing SH or BS at 1 restores the shattering, so the only witness world is that of $[ST \leftarrow 0, BH \leftarrow 0]$: ST returns to default; SH and BS move from **l** to **n**; BH moves from **n** to **f**, since Billy's rock, thrown and no longer preempted, is held at not hitting; BT is unchanged. For Tracks, the only witness world is that of $[F \leftarrow 0, LT \leftarrow 0]$: F returns to default, LT moves from **n** to **f** (the train is held off the left track although the switch is set left), and RT and A move from **l** to **n**. For Doorbells, it is that of $[T \leftarrow 0, L \leftarrow 0]$: T returns to default, L moves from **n** to **f**, R and E from **l** to **n**; M is unchanged. □

If the filter admits the common profile, both switch-settings are causes. If it excludes it, Suzy's throw is not a cause by itself. Computation over all 355 preorders on the four categories confirms that, for componentwise filters, Suzy's throw is a cause exactly when each switch-setting is part of a cause; and in every preorder that leaves Suzy's throw part of a cause only through a conjunction, the conjunction includes Billy's throw, which becomes part of a cause. This is the structure Halpern (2016, 84–85) identified for late preemption; Theorem 1 shows that the escape from it is the switching verdict.

The calibrating cases already force the common profile. In a componentwise filter, genuine prevention requires $\mathbf{l} \succeq \mathbf{n}$, the chain $\mathbf{n} \succeq \mathbf{l}$, and Boulder's only modified witness (the hiker held not ducking as the boulder falls: K from **l** to **f**, H from **n** to **l**) requires $\mathbf{f} \succeq \mathbf{l}$; so $\mathbf{f} \succeq \mathbf{n}$, and the common profile passes. The short circuit adds $\mathbf{u} \not\succeq \mathbf{n}$. Exactly five of the 355 preorders, two of them total preorders, give all four calibrating verdicts, under each definition. For additive filters the witness gains (witness total minus actual total) satisfy

$$g_{\text{Doorbells}} = g_{\text{Tracks}} = g_{\text{Suzy}} = r + a - 2b + c = (r + 2a - 2b) + (c - a) = g_{\text{Chain}} + g_{\text{Boulder}},$$

because a frozen default's change $\mathbf{n} \to \mathbf{f}$ telescopes into $\mathbf{n} \to \mathbf{l}$ and $\mathbf{l} \to \mathbf{f}$. Compensation therefore does not help: any additive score that keeps the chain and the ducking counts both switches.

Theorem 2 (original and updated definitions). Under event coding and the original or updated definition, the flip in Blocked tracks has the witness $[EF \leftarrow 0, LB \leftarrow 0]$, whose profile is that of genuine prevention's witness plus one root moving from deviant to default.

Hence every anonymous Pareto filter that makes the antidote a cause makes the flip a cause of the non-arrival. For additive filters, $g_{\text{Blocked}} = g_{\text{Prevention}} + r$; a componentwise filter admits the flip's witness iff $\mathbf{l} \succeq \mathbf{n}$.

Proof. The antidote's witness $[A \leftarrow 0]$ returns A to default and moves V from **n** to **l**; P is unchanged. In Blocked tracks the actual world is $(EF, LB, RB, TA) = (1, 1, 1, 0)$ and the witness world $(0, 0, 1, 1)$: EF and LB return to default, RB is unchanged, and TA moves from **n** to **l** (the train arrives, as its equation now demands). Sufficiency holds: restoring $EF = 1$, with or without $LB = 0$, gives $TA = 0$, since the right track is blocked. The gains are $r + b - a$ and $2r + b - a$. □

Under the modified definition the flip has no witness of its own, since no blockage may be cleared; Halpern (2015, 8) notes that with that definition "this becomes a non-problem". Strictly, the flip remains part of a cause: the conjunction of the flip and the left blockage has as witness the world in which both are undone, which is the world of Theorem 2, so every anonymous Pareto filter that keeps the antidote admits it (for componentwise filters, exactly when $\mathbf{l} \succeq \mathbf{n}$; computed). Whether this matters depends on whether parts of causes are counted as causes. I set it aside and state the dilemma for single candidates.

Corollary (the dilemma). Under event coding, no anonymous Pareto filter makes the antidote and Suzy's throw causes, leaves Billy's throw out of every cause, and makes neither the switch-setting in Doorbells nor the flip in Tracks or in Blocked tracks a cause, under any of the three definitions.

Under the modified definition this is Theorem 1; under the older definitions it is Theorem 2. An exhaustive check over all 355 componentwise preorders, with candidates of every size and AC3 applied after filtering, finds no exception under any definition; a check of 3,000 random additive scores finds none either (Appendix).

Theorem 1 does not hold for the older definitions, and the reason locates the dilemma. They give Suzy a second witness, $[ST \leftarrow 0, BT \leftarrow 0]$, which resets Billy's throw to its default; every variable in it improves or stays put, so any Pareto filter that keeps the chain admits it. The switch-settings in Tracks and Doorbells have no such witness: in Doorbells, cancelling the press violates sufficiency. So the older definitions can exclude the switching witnesses while keeping Suzy. The score $(r, a, b, c, d) = (2, 0, -1, -5, -3)$ does exactly this under both older definitions and also keeps prevention, the chain and the ducking. But it counts the flip in Blocked tracks, and in Halpern's model with clear tracks (via $[F \leftarrow 0, LB \leftarrow 1]$, gain $a - b$).

The witness that rescues Suzy under the older definitions and the witness that convicts the flip are the same kind of move: a deviant root, Billy's throw or a blockage, is reset to its default. The modified definition forbids the move. That disposes of Blocked tracks and, by the same stroke, leaves Suzy only the frozen witness, which has the switching profile. The choice between definitions decides which distinction normality must draw; it does not remove the need for one.

5. Pricing the non-anonymous escapes

The theorists in question do not propose an anonymous policy. Halpern and Hitchcock call their rules rough-and-ready, allow cross-variable comparisons where "such a comparison clearly seems warranted" (2015, 32), and disclaim the task of determining the right normality ordering (26 fn. 13). The dilemma is therefore not a refutation. It instead identifies a limit on a specified class of transferable policies:

(P1) Under event coding, every anonymous Pareto filter that makes the antidote and Suzy's throw causes, and leaves Billy's throw out, makes some switch a cause, under each HP definition (Corollary).

(P2) Maintain event coding for the variables involved: throws, hits, rings, barks and fresh blockages are occurrences on the codings used here. This is an explicit assumption; the inertial alternative for standing blockages is considered below.

(P3) The antidote and Suzy's throw are causes; Billy's throw and the switches are not.

(C) So any normality ordering that respects (P3) under an HP definition either violates Pareto improvement, which is Halpern and Hitchcock's first guideline, or ranks two witnesses with the same profile differently, that is, by what their variables represent. Under the modified definition only the second option is open, since Theorem 1 does not use Pareto.

The conclusion is not that such orderings are illegitimate, but that their verdicts on switches are only as good as the mechanism-specific judgment they encode. The rest of this section examines one such judgment and the limits of a published motivation for it.

One additive separation. Hold the root and other category contributions fixed, and allow only the frozen-default score to depend on the mechanism. With scores c_{BH} and c_L for the frozen hit and bell, witness gains differ by $c_{BH} - c_L$. Within this restricted repair, keeping Suzy and excluding Doorbells requires $c_{BH} > c_L$, and likewise for the train held off the left track. General non-anonymous orderings could instead distinguish root changes, other transitions or interactions; Theorem 1 does not force them to locate the difference in frozen mechanisms. This escape is formally cheap. With frozen bell and left track scored -5 and every other frozen default 0 (and $r = 2$, $a = 0$, $b = -1$, $d = -3$), the modified definition counts the antidote, the chain's start, the ducking and Suzy's throw as causes, and leaves Billy's throw, the flip in Tracks and the switch-setting in Doorbells out of every cause. It is also Halpern's own judgment on Tracks: "the world where the train does not go on the left track despite the switch being set to the left is less normal than the actual world" (2015, 8). In componentwise terms, the escape indexes the preorder by variable: $\mathbf{f} \not\succeq \mathbf{n}$ for the track and the left bell, while $\mathbf{f} \succeq \mathbf{n}$ for Billy's hit and the ducking. The question is what grounds the difference.

The reliability asymmetry. One published ground for distinguishing frozen mechanisms is reliability. Reporting Beckers and Vennekens (2018), Xiao and Xiong (2026, 8) write that "the key distinction between the early preemption and the switch example lies in whether we take into account the possibility that the backup mechanism malfunctions"; they add that "it is quite plausible to regard the supervisor's shooting accuracy as far more uncertain than the breakage of a sturdy track". On this view a frozen default is abnormal to the degree that its mechanism is reliable. That fits the textbook cases, where the preempted backup is an agent and the switch a sturdy device, but it makes the verdict depend on how reliable the backup was rather than on whether it was preempted. Two tests bring this out.

> Sprinkler. A fire is put out by a firefighter's hose; a sprinkler system with a failure rate no higher than a doorbell circuit's would have put it out moments later.

Structurally this is late preemption, with the sprinkler's activation for Billy's throw. Within the additive repair just specified, if the frozen sprinkler is as abnormal as the frozen bell and the other contributions agree, the firefighter is not a cause by itself under the modified definition; on computation, the firefighter is part of a cause only together with the sprinkler's activation, which is itself part of a cause. Most will say the firefighter put out the fire, however reliable the backup.

Seedlings. $P = 1$ (a field is sown), $S = 1$ (a locust swarm arrives); it is a dry week and no rain is possible. $G = P \wedge S$ (the swarm strips the seedlings), $D = P \wedge \neg S$ (they die of drought), $X = G \vee D$ (they are destroyed). Actually $G = 1$, $D = 0$, $X = 1$.

The map $P \mapsto M$, $S \mapsto T$, $G \mapsto R$, $D \mapsto L$, $X \mapsto E$ is an isomorphism onto Doorbells that preserves equations, actual values and event-coded defaults. So any account whose verdicts depend only on those features treats the swarm as it treats the switch-setting. A repair varying only frozen-mechanism reliability could separate them only if unwatered seedlings in a dry week were less certain to die than a pressed, left-switched bell is to ring. If they are not, it denies that the swarm is even part of a cause of the destruction (computed). A defender may reply that the coarse outcome hides timing: with destruction dated, the swarm is a but-for cause of destruction on the third day. That is a reasonable repair, but then representation, not normality, separates the swarm from the switch, and Doorbells admits no corresponding repair, since the bark comes at the same time either way.

Gallow's own model of Doorbells takes a representational route: it has one switch variable whose values encode both setting and signal, and his negative verdict holds "so long as the switch's directing a signal to the right is no more deviant than its directing a signal to the left" (2021, 86). That is a comparison among the values of a single variable, which no anonymous filter on a binary route model can express. Whether such representations can be selected by a principle stated before the verdicts is a question about model aptness. Blanchard and Schaffer (2017), as Andreas and Günther (2024, 27–28) report, argue that apt causal models make a default–deviant distinction unnecessary. I do not settle that question here.

Inertial coding of standing blockages. Under the older definitions, Theorem 2 has a natural escape. If a standing blockage is a default, inertial state, in the sense in which Gallow (2021, 86–87) contrasts "deviant, noninertial events" with "default, inertial states", then clearing it moves a root from default to deviant. Componentwise filters then exclude the flip's witness, and its additive gain falls to $b - a = g_{\text{Prevention}} - r$, which can be negative while the antidote's is not (computed for all five calibrated preorders). The cost is that the escape answers a question the verdict does not depend on. Let both tracks have been blocked a minute ago by landslides, which are occurrences. The coding is then the event coding of Theorem 2, and the flip counts again. Whether the flip caused the non-arrival should not depend on how long the tracks have been blocked. The escape also leaves Theorem 1 untouched.

6. Exceptionless conformity

Halpern and Hitchcock note that their treatments of the short circuit and of preemption involve exceptions to the default that endogenous variables obey their equations (2015, 36). They show how to remove the short circuit's exception by adding a variable for the second bodyguard's intentions, and they conjecture "that it is always possible to formally construct additional variables to do this" (37). Under the modified definition, carrying this programme to completion has a sharp cost.

Lemma (frozen witnesses). In a deterministic acyclic model with fixed context and non-intervened roots fixed, under the modified definition, let $\vec{X} = \vec{x}$ be a cause of φ with a witness whose world violates no non-root equation. Then every member of $\vec{X}$ is a root, and φ depends counterfactually on $\vec{X} = \vec{x}$.

Proof. Let $\vec{R}$ be the root members of $\vec{X}$. Since the witness world violates no non-root equation and every root outside $\vec{R}$ keeps its actual value, acyclicity and uniqueness of solutions make it identical to the world under $[\vec{R} \leftarrow \vec{r}']$ alone. Since φ fails there, $\vec{R}$ is nonempty and satisfies the witness condition with nothing frozen; the same world passes any filter that the original witness passed. If $\vec{X}$ had a non-root member, $\vec{R}$ would be a proper subset satisfying AC1 and AC2, contrary to AC3. □

The argument adapts the elimination-and-minimality reasoning of Halpern's (2016, 121, 133–134) stability theorem for conservative extensions to a single conforming witness. A check on 400 random models found no counterexample.

Suppose, then, that conformity is made exceptionless: every equation violation is strictly less typical, and no special comparison compensates for it. Since the actual world violates no equation, an admissible witness world violates none, and by the Lemma only root but-for causes, possibly conjunctive, survive. The ducking in Boulder is a non-root and is lost; so is the chain's intermediate $I = 1$; in late preemption the throws become jointly a cause, with Billy's throw part of it (all computed). The conjecture may still be true as a claim about representing orderings. What fails is the hope that equation-conformity, once all exceptions are explained away, separates preemption from switching under the modified definition.

Xiao and Xiong's (2026, Definition 13, 6) refinement of Hitchcock's active-route account requires a witness with off-route intermediates fixed at their actual values to be "available": it "does not violate the structural equations of the model" (Definition 12, 5). They prove that their causes are modified-HP causes (Proposition 1, 6). If availability is evaluated in a deterministic acyclic model at the same fixed context, with nonintervened roots unchanged, the Lemma applies: only root but-for causes survive. Their acknowledged difficulty with the trainee's firing in early preemption (8) is consistent with this consequence. If availability instead permits changes in background or exogenous mechanism variables, the fixed-context premise fails, and the Lemma alone does not establish the consequence. Their proposed use of nondeterministic models and variables for accuracy and nerve therefore needs a separate analysis. The theorem here concerns the explicit fixed-context reading, not every possible interpretation of availability.

7. A parallel in regularity theory

Beckers (2026, 2) argues that Andreas and Günther's accounts face a dilemma of related shape. Their final theory excludes the flip in the simple switch by structural conditions: condition (5) keeps the regularities of the candidate's descendants, and then no admissible set of facts satisfies condition (2) (Andreas and Günther 2024, 25). Beckers reports that their 2024 accounts need the deviancy condition to give the same verdict on two alternative switch models, one of them from Hitchcock (2009), and he argues that the deviancy condition then gives the wrong verdict on early preemption in which the backup process is not deviant (Beckers 2026, 2, 10–11). A drought preempted by locusts is such a case: on my computation their final theory counts the swarm a cause without the deviancy condition and not with it, since the normal absence of rain must be left out of the facts. The same computation shows that their other conditions already exclude both the Doorbells switch-setting and the Seedlings swarm. Since the deviancy condition only removes causes, their theory excludes the Seedlings swarm whatever deviancy is assigned; for them, Seedlings bears on the structural conditions, not on deviancy. Section 4 shows that normality-augmented HP faces a counterpart of Beckers's dilemma, and that changing the definition only moves it.

8. Objections and replies

Objection 1: aptness. Tracks and Doorbells are route models. Halpern's three-variable switch model gets the switch right without normality under all three definitions (2015, 7).

Reply. Theorem 2 arises in the blockage representation that Halpern (2015, 7) uses. Theorem 1 needs route variables, and in Doorbells they are hard to avoid: each bell's ringing is a distinct event that can itself be an effect, and Gallow's own model keeps the bell variables, routing the signal through a single many-valued switch variable with an explicit normality comparison among its values (2021, 85–86). A theorist who rejects route models owes a rule, stated before the verdicts, that selects the non-route model for switches and the route model for late preemption, whose route variables (SH, BH) nobody proposes to remove.

Objection 2: the coding. Event coding does the work; another coding might escape.

Reply. Theorem 1 needs only that throws, settings, hits, rings, shattering and barking be deviant and their absence default, which the codings used for these cases in this literature grant. Theorem 2 needs blockages to be deviant; the inertial alternative was priced in Section 5.

Objection 3: graded causation. Halpern's graded account keeps causes with abnormal witnesses and compares their quality (2016, §3.3).

Reply. The theorems concern the filter. If a graded account compares only relative witness quality through the same anonymous transition profile, it assigns these witnesses equal quality. That conditional parity does not cover graded accounts that use absolute normality levels, mechanism-specific comparisons or other information.

Objection 4: the verdicts. Some deny that switches are non-causes, or doubt the contrast with preemption; Hitchcock (2009), as Beckers (2026, 10) reports, presented a switch model to challenge it.

Reply. Then the dilemma has no grip, and normality has no switching work to do. The result is conditional on the verdicts in Section 3, which the normality approach was introduced to respect.

9. Conclusion

Under event coding, an anonymous Pareto filter cannot keep genuine prevention and late preemption while excluding the standard switches under any of the three HP definitions. The modified definition equates Suzy's sole witness profile with the switching profiles; the older definitions admit a blocked-tracks witness that improves on genuine prevention's. This is a conditional limitation on transferable policies, alongside Wysocki's stronger obstruction for conjoined cases. A separating policy may use principled information beyond the profile. Varying only frozen-mechanism scores exposes a reliability dependence; it is one escape, not the only one. Exceptionless equation-conformity has a separate fixed-context cost: only root but-for causes remain. The open issue is which additional normality information or model-selection principle justifies the desired distinctions.

Appendix. Proofs and computation

Witness uniqueness and AC3. In genuine prevention the only witness world is $(P, A, V) = (1, 0, 1)$. In the chain, freezing I or Y at 1 restores $Y = 1$, and setting either to 0 violates sufficiency, so the only witness is $[C \leftarrow 0]$. In Boulder, the modified definition admits only $(F, B, K, H) = (1, 1, 0, 1)$; the older definitions also admit $(0, 1, 0, 1)$, in which B moves from **l** to **u**, and the calibration conditions are unchanged. In the short circuit,

the only witness is $[A \leftarrow 0, P \leftarrow 1]$, which moves P from **l** to **u** and V from **n** to **l**. For a single candidate, AC3 holds automatically, so Theorems 1 and 2 need no minimality argument; a single candidate is a cause iff some witness is admitted. The corollary's claim that Billy's throw is in no cause, and the remarks on conjunctive causes, rest on the exhaustive computation. Under the older definitions the switches have no further witness worlds: every other contingency that removes the outcome violates sufficiency. In Doorbells, cancelling the press leaves the barking absent when the switch is restored; in Tracks, the other contingencies stop the arrival only by holding A or RT at 0, which also stops it when the switch is restored.

Computation. A Python implementation of the three definitions (Halpern 2015, 3–4), with the normality filter applied before AC3, enumerates every candidate subset of every model above, including those containing the outcome, and every admissible witness. It checks Theorems 1 and 2 and the corollary for all 355 preorders on the four categories, the gain identities symbolically, 3,000 random integer additive scores with $r > 0$, the boundary score, the escapes of Section 5, the Lemma on 400 random acyclic binary models (357 causes, 940 conforming witnesses), and Andreas and Günther's (2024) conditions with and without deviancy. Without normality the implementation reproduces Halpern's (2015) verdicts on the switch models and the short circuit.

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